Date	1/10/2026	Maximum Marks	4 Marks
Team ID	SWTID-2026-4326		
Project Name	Streamlining IT Procurement automating laptop orders with flow designer		

Project Design Phase-II

Technology Stack (Architecture & Stack)

Technical Architecture:

The architecture below illustrates the end-to-end automated IT procurement system, mapping client interfaces, ServiceNow Flow Designer engine, cloud databases, and external enterprise API integrations.

Architecture Guidelines Compliance:

- Includes application logic blocks (Service Portal, Flow Designer, Integration Hub).
- Provides clear infrastructural demarcation between Local Client Endpoints and Cloud SaaS/PaaS platforms.
- Indicates external third-party interfaces (HRIS, Courier Webhooks, Vendor ERP).
- Includes cloud database storage (CMDB, Transactional DB) and predictive ML models.

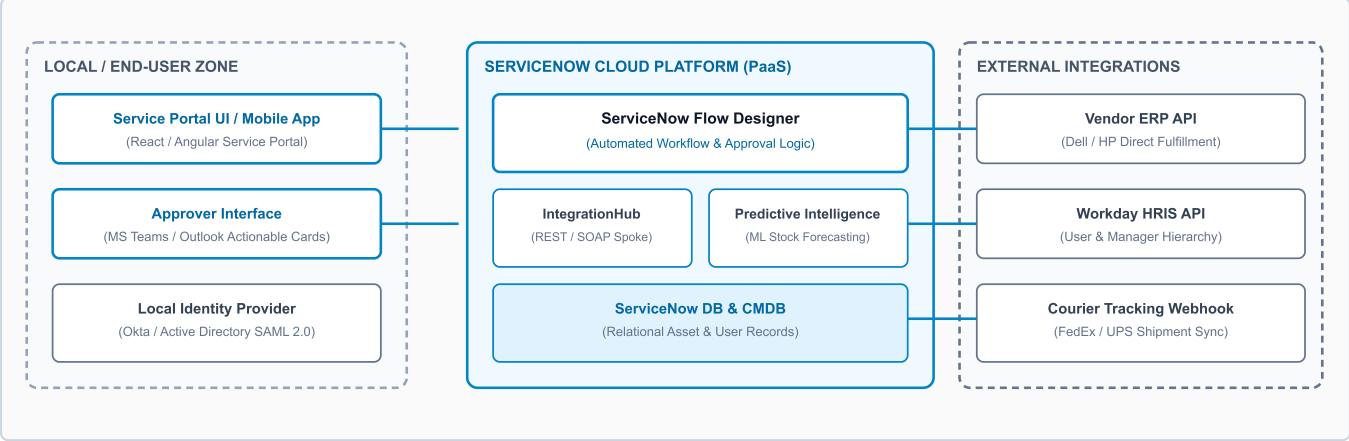


Table-1: Architecture Components

S.No	Component	Description	Technology
1.	**User Interface**	End-user self-service web catalog and manager approval dashboard interface.	HTML5, CSS3, JavaScript, React.js (Service Portal Framework)
2.	**Application Logic-1**	Flow Designer orchestration engine for order validation, routing, and approvals.	ServiceNow Flow Designer / JavaScript Engine

Table-1: Architecture Components (Continued)

S.No	Component	Description	Technology
3.	**Application Logic-2**	Automated notification engine for MS Teams and email approvals.	IntegrationHub Spokes, Webhook Listeners
4.	**Application Logic-3**	Virtual Agent bot service for order tracking inquiry.	ServiceNow Virtual Agent, NLU Engine
5.	**Database**	Transactional storage for active laptop orders, approval logs, and user sessions.	ServiceNow Relational Database (MariaDB back-end)
6.	**Cloud Database**	Central Configuration Management Database (CMDB) tracking laptop hardware assets.	ServiceNow CMDB Cloud Service
7.	**File Storage**	Storage for procurement invoices, PO PDFs, and user sign-off docs.	ServiceNow Attachment API / Cloud Object Storage
8.	**External API-1**	Direct integration with hardware vendor for automatic PO dispatch.	REST API / JSON Web Services (Dell/HP Vendor Portal)
9.	**External API-2**	HR System integration to verify employee eligibility and reporting manager hierarchy.	Workday / SAP SuccessFactors REST API
10.	**Machine Learning Model**	Predictive demand forecasting model for proactive laptop inventory management.	ServiceNow Predictive Intelligence / Python Scikit-Learn
11.	**Infrastructure (Server / Cloud)**	High-availability cloud PaaS hosting application logic and data pipelines.	ServiceNow Enterprise Cloud Infrastructure (Multi-instance Architecture)

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1.	**Open-Source Frameworks**	Client-side libraries used for responsive portal components and charting.	React.js, AngularJS, Bootstrap 5, D3.js
2.	**Security Implementations**	Role-based access control (RBAC), SSO authentication, data encryption, and OWASP compliance.	OAuth 2.0, SAML 2.0, SHA-256, TLS 1.3 Encryption, IAM Controls
3.	**Scalable Architecture**	Decoupled 3-tier cloud architecture separating UI, workflow engine, and storage layer.	ServiceNow Multi-Instance Platform, REST Microservices
4.	**Availability**	Redundant multi-region datacenter deployment ensuring continuous operational availability.	F5 Load Balancers, Active-Active High Availability Clusters

Table-2: Application Characteristics (Continued)

S.No	Characteristics	Description	Technology
5.	**Performance**	High-speed caching and edge delivery mechanisms ensuring sub-second response times under peak request surges.	Akamai CDN, Redis In-Memory Cache, Asynchronous Workflow Triggers

Architecture Summary & Design Justification

System Integration Strategy:

The selected technology stack directly addresses manual inefficiencies in enterprise laptop procurement. By leveraging **ServiceNow Flow Designer** as the core orchestration middleware, manual handoffs between department managers, IT asset teams, and external hardware vendors are replaced by automated triggers.

Key Technical Highlights:

- **Seamless HR & ERP Connectivity:** Automated validation against Workday HRIS eliminates invalid requests and streamlines manager auto-assignment.
- **Multi-Channel Approval Engine:** Integrates directly with Microsoft Teams and Email actionable messages, cutting approval latency from days to hours.
- **Predictive Inventory Control:** Embedded Machine Learning models forecast department hardware demand based on historical onboarding trends, reducing out-of-stock delays.
- **Enterprise-Grade Security:** Built-in OAuth2 authentication and TLS 1.3 encryption safeguard employee data and corporate purchasing credentials.